

Can Thinking Harder (Without Writing More) Make AI Smarter?

Track: DataForge x Pathway - Explain the Frontier | **Live Explainer:** theshrihari.github.io/Dataforge-Submission/ | **Code:** github.com/TheShriHari/Dataforge-Submission

Topic: Scaling test-time compute via recurrent latent reasoning without emitted tokens

1. Central Claim

"Increasing recurrent latent computation steps at inference time improves reasoning accuracy without producing extra natural-language tokens - but gains diminish and each extra step costs time and compute."

2. Motivation: Latent Iteration vs. Chain-of-Thought (CoT)

Today's models scale answer-time thinking by writing intermediate text steps (Chain-of-Thought; Snell et al., 2024, [arXiv:2408.03314](https://arxiv.org/abs/2408.03314)). However, verbalized CoT has core physical bottlenecks:

- 1. **Memory & Attention Bloat:** Emitted tokens expand the context window, causing $O(N^2)$ self-attention cost and quadratic KV-cache growth.
- 2. **Syntactic Drift:** Conditioning on text means an early mistake permanently corrupts downstream steps.
- 3. **Latent Recurrence:** Instead of writing tokens, the model passes a continuous hidden tensor through a weight-shared core for K steps: $h_k = \text{Core}(h_{k-1}, x)$. Compute scales while memory remains constant ($O(1)$) with zero emitted text tokens.

3. Empirical Findings: 2D Maze Spatial Pathfinding

We evaluated this mechanism using a 123,713-parameter shared ConvGRU core (`m1/model.py`) on 100 held-out 15×15 mazes (Seed 999). Each step expands the receptive field by 2 grid cells:

- **Steep Early Phase ($K = 1 \rightarrow 10$):** Solve rate rises from **1.0%** (0.67 ms) to **44.0%** at $K = 5$ (2.91 ms) and **81.0%** at $K = 10$ (6.16 ms). Mean Path IoU rises from 55.9% to 96.6%, resolving maze connectivity.
- **Diminishing Returns ($K = 10 \rightarrow 20$):** $K = 10 \rightarrow 20$ yields only a **+7.0%** solve gain (81% $\rightarrow$ 88%) while latency doubles (6.16 ms $\rightarrow$ 12.88 ms).
- **Extrapolation:** $K = 16\text{--}20$ extends beyond the $K \sim \text{Uniform}(1, 15)$ training curriculum (`m1/train.py`). The model generalizes because the core is weight-shared with no step-count parameters.

4. Paradigm Comparison

Dimension	Toy Latent Maze Solver	BDH-CQ Foundation Model (Kosowski et al., 2026)	Autoregressive CoT (Snell et al., 2024)
Domain & Scale	2D Mazes (15×15), 123K params	ARC-AGI-1 Reasoning, $\sim 150\text{M}$ params	Symbolic Math & Language, $> 7\text{B}$ params
Effort vs. Accuracy	Solves scale 1.0% ($K = 1$) $\rightarrow$ 88.0% ($K = 20$)	Pass@2 scales 21.0% (Low) $\rightarrow$ 29.5% (High)	Scales with sample count & search beam
Compute & Latency	Linear runtime: 0.67 ms $\rightarrow$ 12.88 ms	Low effort cuts hardware compute by 22%	$O(N^2)$ sequence compute & KV expansion
State Footprint	Constant $O(1)$ tensor ($\mathbb{R}^{48 \times 15 \times 15}$)	Constant latent manifold updates	Dynamic context growth per token
Observability	Continuous probability grid readout	Hidden continuous activations	High: Human-readable natural text

5. Architectural Roles: BDH vs. BDH-CQ

- **The Dragon Hatchling (BDH)** (Kosowski et al., 2025, [arXiv:2509.26507](https://arxiv.org/abs/2509.26507)): Foundational post-transformer architecture with recurrent latent state updates and non-autoregressive parallel processing.
- **BDH-CQ** (Kosowski et al., 2026, [arXiv:2608.09888](https://arxiv.org/abs/2608.09888)): Specializes BDH with in-context demonstration learning ("Context Queries") and LOW/MED/HIGH effort tiers without scratchpad tokens.

6. Evidence Classification & Maturity

- **Advantage Over Incumbents:** Eliminates KV-cache expansion and syntactic fragility; enables variable inference effort in constant memory.
- **Untested Frontiers:** Untested on open-ended generation where language tokens are the necessary output.
- **Evidence Integrity:** Maze numbers are our own empirical measurements (100 mazes, seed 999). BDH-CQ Table 5 results (21.0% $\rightarrow$ 29.5%) are developer-reported by Pathway and not independently reproduced.

7. Primary Limitation & Next Steps

- **Auditability Trade-off:** Latent reasoning lacks an English scratchpad. Diagnosing failure at step $K = 4$ requires probing continuous vectors rather than reading text. Developing zero-overhead diagnostic probes remains an active open challenge.
- **Primary Sources:** Kosowski et al. (2025, [arXiv:2509.26507](https://arxiv.org/abs/2509.26507)); Kosowski et al. (2026, [arXiv:2608.09888](https://arxiv.org/abs/2608.09888)); Snell et al. (2024, [arXiv:2408.03314](https://arxiv.org/abs/2408.03314)).